

Multiple-Input Boost Converter to Minimize Power Losses due to Partial Shading in Photovoltaic Modules

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Abstract— A multiple-input boost (MIB) converter is proposed to implement maximum power point tracking (MPPT) for series strings of solar cells, typically connected across bypass diodes in photovoltaic (PV) modules. The proposed configuration is intended to increase energy harvest from the PV module during instances of partial shading. Results from an experimental prototype demonstrate that the MIB converter extracts up to 9.6 % more power compared to a single-input boost converter in a 120W PV module. The proposed topology can be adopted in emerging distributed MPPT system architectures that utilize microconverters and microinverters.

Keywords- Maximum power point tracking, Microconverter, Microinverter, Multiple-input boost converter, Partial shading, Photovoltaics

I. INTRODUCTION

The penetration of switch-mode power converters in PV energy conversion systems has significantly increased. Conventional installations where large PV arrays are connected to centralized power converters (Fig. 1(a)) are giving way to distributed architectures. In one such emerging architecture, dc-dc converters (referred here as *microconverters*) realize MPPT algorithms for individual PV modules [1]-[2]. A central inverter converts the dc voltage sourced by a series string of such modules to mains-compatible ac voltage (Fig. 1(b)). An alternate architecture couples each PV module with a dedicated dc-ac inverter, commonly called a *microinverter*. Each module produces power at ac-mains voltage, and a central inverter is unnecessary [3]-[4] (Fig. 1(c)). These microconverter and microinverter architectures are commonly referred as distributed maximum power point tracking (DMPPT) systems.

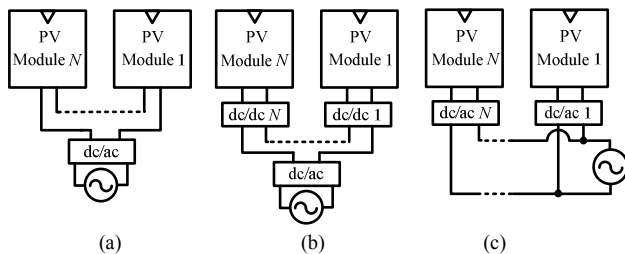


Figure 1. Traditional and emerging PV system architectures. (a) Central inverter, (b) microconverter architecture, (c) microinverter architecture

Distributed maximum power point tracking architectures increase energy harvest by reducing the effects of panel mismatch and partial shading as MPPT is implemented for each PV module. Series strings of modules are hence not current limited by shaded or faulty modules. System reliability is also improved due to the inherent structural redundancy. As a result of these advantages, DMPPT systems are expected to provide lower levelized cost of energy (LCOE) when compared to the conventional central systems.

This work focuses on increasing the energy harvested by DMPPT converters during instances when PV modules are partially shaded. The distributed architectures draw most of their advantages (especially the increased energy harvest due to MPPT implementation for each module) as power converters have direct access to individual PV modules. Taking the trend of ‘distribution’ a step further, a multiple-input boost (MIB) converter topology that executes MPPT separately for sections of an individual PV module is proposed. The MIB converter utilizes series strings of solar cells conventionally connected across bypass diodes as inputs. Figure 2 shows how the MIB topology can be incorporated as a front-end converter to execute MPPT in microconverters or microinverters.

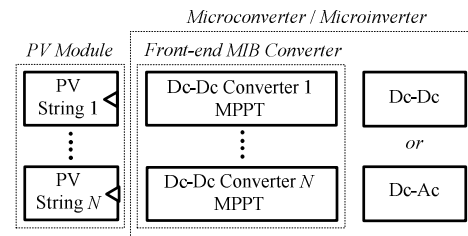


Figure 2. Block diagram of a typical DMPPT converter with MPPT implemented in a front-end MIB converter

The remainder of this paper is organized as follows. Section II details the construction of PV modules and describes the topology of the MIB converter. The impact of partial shading on PV power output is gauged through simulations in Section III. Results from an experimental prototype are described in Section IV. Finally, conclusions and possible future work are highlighted in Section V.

II. PHOTOVOLTAIC MODULE CONSTRUCTION AND MIB CONVERTER TOPOLOGY

The SS-SP-120, 120W PV module is employed in this work. The module parameters are listed in Table I. The physical construction of the module is depicted in Fig. 3 (a). The module is built with 36 solar cells connected in series. The series combination is split into two strings of 18 cells each. Two bypass diodes are connected across each string. These diodes are accessed from the weather-proof junction box mounted at the back of the PV module. To ensure that the MIB converter has access to both strings, the trace connecting the strings is cut, and the bypass diodes are removed. The two strings now form two inputs for the MIB converter. This modification is illustrated in Fig. 3(b).

Symbol	Quantity	Value
V_{OC}	Open-circuit voltage	21 V
I_{SC}	Short-circuit current	7.7 A
V_M	Rated voltage	17 V
I_M	Rated current	7.1 A

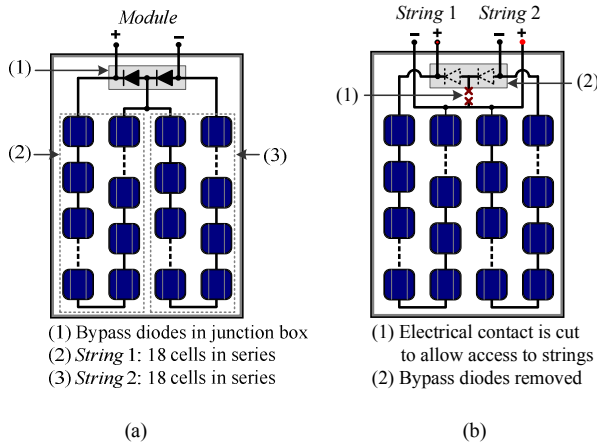


Figure 3. The SS-SP-120 module utilized in this work. (a) Layout of cells in original module, (b) modified module for use with MIB converter

A conventional single-input boost converter to implement MPPT for the SS-SP-120 module is shown in Fig. 4. Module voltage, V_{Module} , and current, I_{Module} , are utilized to execute an MPPT algorithm that determines the switch state, $q(t)$. The proposed multiple-input boost converter that implements MPPT independently for each string is depicted in Fig. 5. In this case, the voltages and currents of each string are utilized to independently implement MPPT. As a matter of notation, MPPT executed with a single-input converter for the entire module is referred as module-level MPPT. Similarly, MPPT executed independently for multiple strings is referred as string-level MPPT. A general N -input MIB topology can be applied in other PV modules (with a different number of bypass diodes).

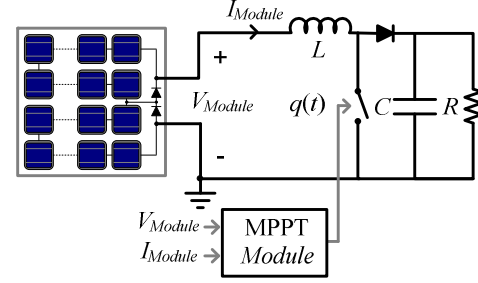


Figure 4. Single-input boost converter for PV module

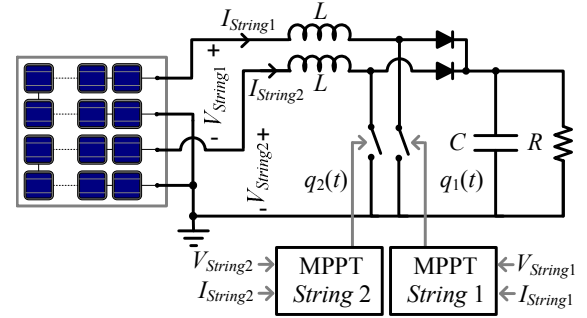


Figure 5. MIB converter to implement string-level MPPT

III. EFFECTS OF PARTIAL SHADING ON PV POWER OUTPUT

As shown in Fig. 3, crystalline-silicon PV modules are built with bypass diodes connected across a specific number of cells (typically across every 18 cells). When the module is uniformly illuminated, the bypass diodes are off. If cells in a string are shaded, the corresponding bypass diode may turn on to conduct the current sourced by non-shaded cells, which reduces the energy harvested from the module. Additionally, under such conditions, local maxima appear in the PV power-voltage characteristic. Commonly used MPPT techniques such as perturb-and-observe, incremental conductance, ripple correlation control, fractional open-circuit voltage, and fractional short-circuit current methods cannot differentiate between local and global maxima (as adopted widely, and suggested originally). For a detailed description of these techniques, refer [5]. Advanced algorithms are necessary to track the global maximum and address this problem. Some algorithms and circuit techniques can be found in [6]-[8].

The simulation results outlined below illustrate the appearance of local maxima and the loss in power due to partial shading. Figure 6 depicts the simulation setup. In all cases, String 1 is illuminated at 1000 W/m^2 . In Case (i), String 2 (refer Fig. 3(b)) is illuminated at the nominal 1000 W/m^2 . For Cases (ii) and (iii) (Figs. 8-9, respectively), the illumination is reduced to 500 W/m^2 and 250 W/m^2 , respectively. In the nominal case, the sum of the powers harvested from the two strings is equal to the power extracted from the module. When String 2 is illuminated at 500 W/m^2 (Case (ii)) the converter can extract 65 W at best, but only 52 W if the MPPT algorithm

tracks the local maximum. Similarly, in Case (iii), a single-input converter can harvest 52 W at best, but only 33 W if the local maximum is tracked. With a multiple-input converter, 91 W and 74 W can be harvested for the same cases described above. This is because, the net power harvested is the sum of the MPP's of the two strings.

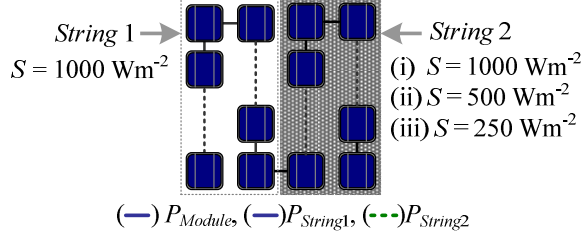


Figure 6. Simulation setup

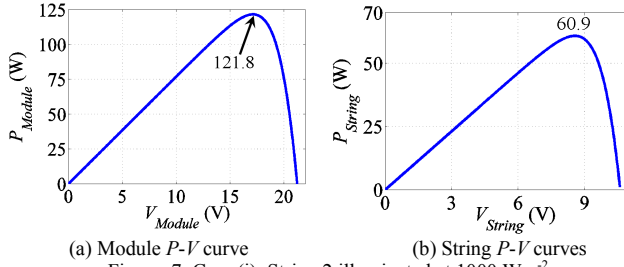


Figure 7. Case (i): String 2 illuminated at 1000 Wm⁻²

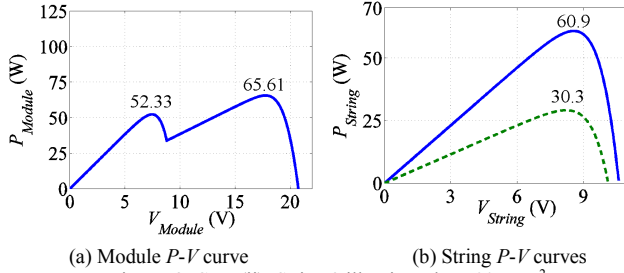


Figure 8. Case (ii): String 2 illuminated at 500 Wm⁻²

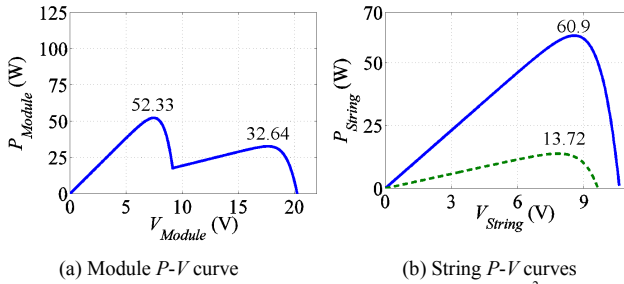


Figure 9. Case (iii): String 2 illuminated at 250 Wm⁻²

The discussion presented in this section suggests that string-level MPPT is a promising method to increase energy harvest during partial shading. This motivates the formulation of the multiple-input boost converter topology to implement MPPT for strings of solar cells in PV modules.

IV. EXPERIMENTAL PROTOTYPE

An experimental prototype of the dual-input boost converter has been built and tested. In particular, the goal of the experimental studies is to gauge the increase in energy harvest during partial shading. The parameters of the MIB converter are listed in Table II.

Symbol	Quantity	Value
L_1, L_2	Input-filter inductances	305 μ H
C	Output-filter capacitance	40.7 μ F
R	Output load	25 Ω
f	Switching frequency	160 kHz

The perturb and observe technique is adopted for MPPT. This well-known technique involves making a small positive (δD) or negative perturbation ($-\delta D$) to the duty cycle of the active switch (D_k) in the boost converter based on how the sign of a previous perturbation affected the average power drawn (P_k) from the PV source. The algorithm is captured by the expressions in (1). The change in duty cycle and power at the k^{th} instant are denoted as ΔD_k and ΔP_k , respectively. For the multiple-input converter, two instances of the MPPT algorithm are simultaneously executed.

$$\begin{aligned}
 \Delta D_k > 0 \quad \Delta P_k > 0 &\Rightarrow D_{k+1} = D_k + \delta D \\
 \Delta D_k > 0 \quad \Delta P_k < 0 &\Rightarrow D_{k+1} = D_k - \delta D \\
 \Delta D_k < 0 \quad \Delta P_k < 0 &\Rightarrow D_{k+1} = D_k + \delta D \\
 \Delta D_k < 0 \quad \Delta P_k > 0 &\Rightarrow D_{k+1} = D_k - \delta D
 \end{aligned} \tag{1}$$

To illustrate improved energy harvest with the MIB converter under partially-shaded conditions, a single cell in String 1 (refer Fig. 3) of the PV module is shaded (as shown in Fig. 10). The power draw by a single-input boost converter and the proposed multiple-input boost converter are recorded. The PV-module voltage, current and PWM signal in the conventional boost converter are shown in Fig. 11. The string voltages, currents and PWM signals for the MIB converter are shown in Fig. 12. The MIB extracts 43.74 W, while the conventional boost converter extracts 32.22 W from the PV module. Based on the 120 W module rating, this translates to an improvement of 9.6 %.

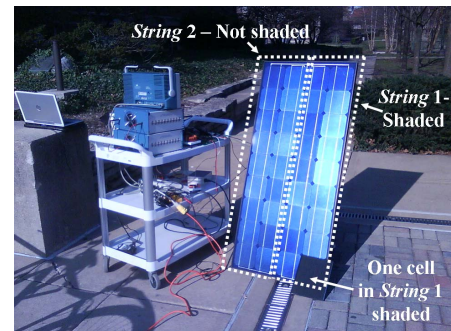


Figure 10. Experimental setup depicting shading condition investigated; $S=750\text{Wm}^{-2}$, $T=14^\circ\text{C}$

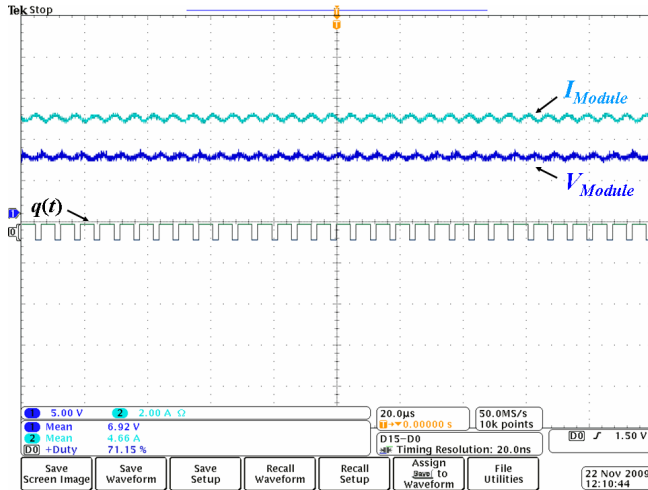


Figure 11. PV module current and voltage with shading

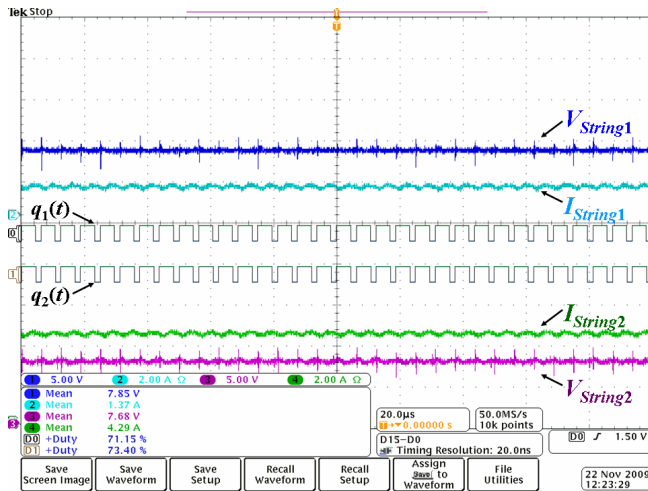


Figure 12. PV strings voltages and currents with shading

V. CONCLUSIONS

A general multiple-input boost converter topology is suggested to execute MPPT in emerging module integrated microconverters and microinverters. The proposed topology increases energy harvest when the module is partially shaded. Results from an experimental prototype demonstrate the benefits of the topology. Future research may investigate multiple input buck or buck-boost topologies for the same application.

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